Assignment 2 Multi-Client Chat Application using TCP and UDP Sockets

Deadline: 06/03/26

A university maintains student records in a CSV file (students.csv). Develop client-server applications in **C** using socket programming that allow students to retrieve academic information from the server.

1(a). TCP Client–Server Application: Retrieve Complete Student Academic Profile

Develop a **TCP-based client-server application** in C with the following requirements:

1. The server reads student records from a CSV file (students.csv) containing the following fields:
    - Roll Number
    - Name
    - Department
    - Semester
    - CGPA
2. The server listens for incoming client connections and handles **multiple clients concurrently** using:
    - POSIX threads (pthread).
3. The client prompts the user to enter a **Roll Number** and sends it to the server.
4. The server searches the CSV file for the requested Roll Number.
5. If the record is found, the server sends the complete student details to the client.
6. If the record is not found, the server sends the message:

Student Record Not Found

7. The client displays the received information and closes the connection gracefully.

1(b). UDP Client–Server Application: Verify Student Record

Develop a **UDP-based client-server application** in C with the following requirements:

1. The server reads student records from the same CSV file (students.csv).
2. The client enters a **Roll Number** and sends it to the server using a UDP datagram.
3. The server searches the CSV file for the requested Roll Number.

4. If the record exists, the server sends back only the following information:
    - Roll Number
    - Student Name
5. If the record does not exist, the server sends the message:

Student Record Not Found

6. The client displays the received response and terminates.

1c. Compare the two implementations based on Connection, Reliability, Ordering, Error Recovery and Communication.